Climate Shocks on Chilean Copper Mining

ENSO-attribution, Impact, and Adaptation

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Research Question

Research question

Does Chilean copper mining sector react **differently** to climate shocks **induced by ENSO**?

The question is broad, so we break it into **three probes**:

1 · Attribution

Are the climate shocks reaching Chile's copper mines genuinely **ENSO-driven**, or merely the background extremes of an already harsh environment?

2 · Impact

How much do those shocks **reduce copper output**, and through which **channels**?

3 · Response

How does the industry **adjust**?

*The hazard returns cyclically, the asset cannot move away from it, and the only margin left is to **adapt in place**.*

Motivation

Research question

Does Chilean copper mining sector react **differently** to climate shocks **induced by ENSO**

ENSO — a **recurring** fluctuation in sea-surface temperatures and atmospheric pressure across the equatorial Pacific that swings exposed regions between wetter and drier extremes. It takes **three states**:

El Niño (warm phase)
Pacific *warms* ⇒ **floods** the central cordillera

Neutral
neither extreme — baseline conditions

La Niña (cool phase)
Pacific *cools* ⇒ deepens **drought** in the north

- **Chile** — a **unique geography**: one of the world's **longest north–south coastlines** (∼4,000 km), Andes to Pacific, hyper-arid Atacama north to a rain-fed temperate centre.
- **Copper** — anchors the economy: ≈**10% of GDP**, >**half of exports**; Chile alone supplies ≈**a quarter** of the world's copper.
- **Water** — the **link**: copper mining is water-intensive and the largest mines sit in the driest north — exactly where ENSO drought bites.

Related Literature

New climate–economy

Climate's effect from year-to-year weather variation (panel FE).

▶ **We:** estimate at the **mine cell**; operational-stress flags, not continuous T/P.

Dell et al. 2014 · Kolstad & Moore 2018 · Mendelsohn et al. 1994

ENSO macro-finance

ENSO linked to output, sovereign/bank risk, prices — one *global* signal.

▶ **We: localise & attribute** — phase × teleconnection × anomaly must align.

Liu et al. 2023 · Damette et al. 2024/25 · Atems & Sardar 2021 · Ubilava 2017

Shocks → trade & firms

Weather shocks cut exports, growth — damage *from output decline*.

▶ **We:** one **export-critical sector at the asset**; damage need not show in output.

Jones & Olken 2010 · Nedoncelle & Wolfersberger 2025 · Cevik & Jalles 2023 · Luo et al. 2026

Adaptation

Damages shrink as agents adjust — adaptation inferred *indirectly*.

▶ **We:** observe it **directly** (desal switch); **document** absorption, not infer it.

Barreca et al. 2016 · Burke & Emerick 2016 · Mendelsohn et al. 1994

Our contribution. The first **ex-post, mine-level, ENSO-attributed** causal panel for Chilean copper output. We attribute operational climate stress to ENSO at the **individual mine**, and read the sector's response off the **adaptation margin directly** — firm accounts and the structural switch to desalinated seawater — not off the output decline alone.

Data Sources

Climate sources

- **CR2MET v2.5** — daily precipitation & max temperature (1960–2021)
- **ERA5-Land** — extends the series (2022–2025)
- **NOAA ONI** — ENSO state; teleconnection ρ, anomaly Z
- **SPEI-12** — 12-month drought index
- **ENSO forecasts:** NOAA CPC + IRI

Bias-corrected to each mine's 0.05° grid cell.

Government sources

- **Copper production:** COCHILCO — monthly output, 19 mines ≈95% and Copper price
- **Water use:** COCHILCO *Consumo de Agua*
- **Water governance:** DGA — rights, aquifer restrictions, scarcity decrees
- **Permits:** e-SEIA filings

Private sources

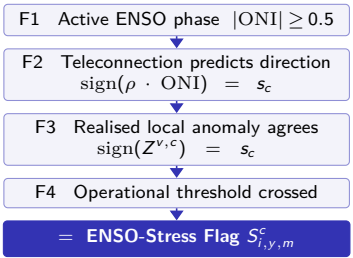
- **Annual statements** — *Memorias Anuales* (CMF)
- **Financial reports** — revenue, capex
- **Cost disclosures** — C1 cash cost, IAS 37 provisions
- **Desalination registry:** ACADES — commissioning dates

Firm water-use microdata (COCHILCO EMPAE) are confidential — desalination commissioning dates proxy the water switch.

Identification — separating ENSO stress from background aridity



The north crosses heat and drought thresholds in *ordinary* years — a threshold crossing alone cannot separate climate stress from baseline aridity. We isolate the ENSO-attributable share with a **4-filter AND-gate**.



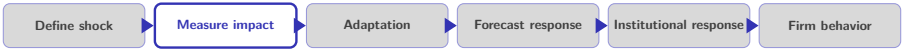
Operational thresholds (Table 1)

Channel	Threshold
FLOOD regional mean	$P_{r,m} \geq 20$ mm/month
FLOOD mine-cell	any day $P_{i,d} \geq 20$ mm
HEAT mine-cell	$T^{\max}_{i,m} \geq 23.5^{\circ}$C
DROUGHT mine-cell	SPEI-12 ≤ -1

Two shock definitions

- **General shock:** *any* threshold crossing.
- **ENSO-attributed:** threshold $\times$ active, directionally consistent ENSO gate. *Intentionally conservative.*

Three estimating equations



Acute channels (FLOOD, HEAT) — mine-month frequency

$$\log(\text{Output}_{\text{kilotonnes}})_{i,t} = \alpha_i + \gamma_{r,t} + \delta_t + \beta\, X^{c}_{i,t} + \theta \log(\text{price})_t + \varepsilon_{i,t}$$

kt: monthly fine-copper output (kilotonnes), in logs. α_i mine scale · $\gamma_{r,t}$ region × calendar-month seasonality · δ_t year · $X^{c}_{i,t}$ is the general-shock (threshold) or ENSO-attributed flag.

Slow DROUGHT channel — mine-year frequency

$$\log(\text{Output}_{\text{kilotonnes}})_{i,t} = \alpha_i + \delta_t + \beta\, X^{DR}_{i,t} + \varepsilon_{i,t}, \qquad X^{DR}_{i,t} = \text{3-yr cumulative SPEI-12 exposure share } (t-1 \text{ to } t-3)$$

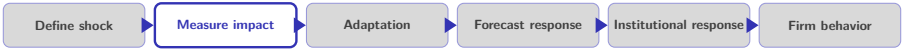
Annual frequency because water scarcity is a *stock*: SPEI-12 already summarises the prior 12 months; output responds through accumulated scarcity. Price omitted (collinear with δ_t).

Adaptation — the moderated interaction

$$\log(\text{Output}_{\text{kilotonnes}})_{i,t} = \alpha_i + \gamma_{r,t} + \delta_t + \beta_1\, \text{Stress}^{c}_{i,t} + \beta_2\, (\text{Stress}^{c}_{i,t} \times \text{Adapt}_{i,t}) + \beta_3\, \text{Adapt}_{i,t} + \theta \log(\text{price})_t + \varepsilon_{i,t}$$

β_1 shock on non-adapted mine · $\boldsymbol{\beta_2}$ **moderation** (cushioning $\Rightarrow$ $\beta_2 > 0$; adapted effect $= \beta_1 + \beta_2$) · β_3 adapted-vs-not level gap in normal months. Inference: **7 regional clusters** $\Rightarrow$ CR0 + restricted wild-cluster bootstrap-t (WCB; Webb weights for the interaction).

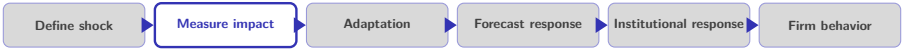
The output impact of climate shocks



Specification	$\hat{\beta}$	SE	p_{CR0}	p_{WCB}	N
***FLOOD**, regional-mean monthly*					
General flood shocks	**−0.047**	0.010	0.002***	0.013**	4,393
ENSO-attributed flood shocks	−0.025	0.020	0.246	0.287	4,393
***FLOOD**, mine-cell daily*					
General flood shocks	−0.018	0.016	0.301	0.395	4,393
ENSO-attributed flood shocks	−0.002	0.028	0.939	0.978	4,393
***HEAT**, mine-cell monthly*					
General heat shocks	−0.003	0.009	0.713	0.686	4,393
ENSO-attributed heat shocks	−0.018	0.009	0.087	0.700	4,393

- **FLOOD** is the **clean signal**: a wet regional month lowers output ∼**4.7%** ($p_{\mathrm{WCB}} = 0.013$); leave-one-region range stays entirely negative. ENSO subset same sign, less precise.
- **HEAT** is a **robust null** at monthly frequency — northern mines operate around routine heat exposure (the apparent ENSO-attributed signal vanishes under the wild-cluster bootstrap).

Drought is cumulative — visible only as ENSO-consistent persistence



Specification (mine-year, $N = 369$)	$\hat{\beta}$	SE	p_{CR0}	p_{WCB}	N
General drought shocks, SPEI-12	+0.075	0.108	0.510	0.752	369
ENSO-attributed drought shocks, SPEI-12	-0.277	0.094	0.025**	0.013**	369

- General drought shocks share: **null** (+0.075). The impact on output concentrates in **ENSO-attributed** persistent drought: −0.277 ⇒ a +10pp exposure share ≈ **2.8% lower** annual output.
- Real but **bounded**: survives gate checks, a lead placebo and every leave-one-region refit — but is identified from the **post-2020 megadrought**.

Why the panel interaction can't pin down adaptation

Run the moderated equation per channel. Desalination should cushion **DROUGHT** ($\beta_2 > 0$), but has **no mechanism** for **FLOOD**/**HEAT** — so those are **built-in placebos**.

Channel (ENSO-attributed)	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	Expectation for β_2
DROUGHT meteo (cum. 3-yr)	—	+0.09 (p=0.90)	—	mechanism: $\beta_2 > 0$
FLOOD region-mean	−0.02	−0.25 (p=0.09)	+0.37	placebo: $\beta_2 \approx 0$
FLOOD mine-cell	+0.02	−0.14 (p=0.49)	+0.36	placebo: $\beta_2 \approx 0$
HEAT mine-cell	−0.02	+0.02 (p=0.88)	+0.36	placebo: $\beta_2 \approx 0$

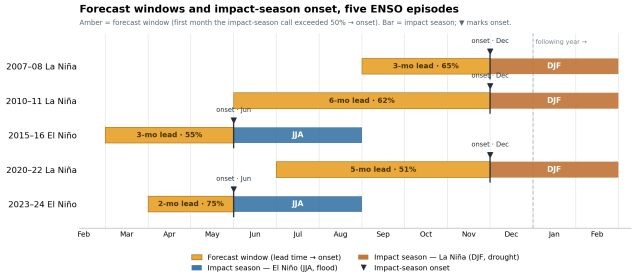
- The *mechanism* channel **DROUGHT** is a **clean null**; the only marginally significant coefficient is the **FLOOD region-mean placebo** — not even marginal under WCB (p=0.09 → 0.23) and **fragile** (dropping one mine moves p from 0.003 to 0.50).
- β_2 tracks **mine type** (adapted mines sit in the driest north), not the water switch — Adapt proxies type because firm water use is confidential.

Conclusion

Panel interaction **cannot identify adaptation** ⇒ compare the **same mine, before vs. after** (event study) and read adaptation off the firm + institutional record.

Do mines respond to ENSO forecasts? — reading the signal

NOAA/IRI publish a **consensus probabilistic** ENSO forecast each month: for every upcoming three-month season, a probability for each phase.



We read it against the impact season, not the $|0.5|$ ONI crossing. Onset timing is hard through the spring predictability barrier; the mine acts on the season its operations are at risk:

- flood — austral winter (JJA), cordillera mines;
- drought — austral summer (DJF), northern mines.

Forecast window: first month the probability exceeds **50%** for the phase of interest (▼).

Key takeaway

A majority-probability signal was present in **all five episodes** — 2 to 6 months ahead.

Do mines respond to ENSO forecasts?

Define shock → Measure impact → Adaptation → **Forecast response** → Institutional response → Firm behavior

Three observable behaviors — two firm-level, one regulatory — and none shows a response.

Register	If anticipating...	Finding
Monthly output	Ramp into forecast window	Every episode's output ramp falls within its neutral-year ± 1 s.d. band.
e-SEIA permits	Water-related filings in window	**Zero** water/flood-adaptation permits across all windows. *Desal clusters post-2022 Water Code.*
DGA decrees	Cluster ahead of dry season	47% issued in-season (Dec–May); only 27% in forecast window. *Signals scarcity, not anticipating.*

Key finding

Adaptation is **long-term and structural**, not a short-run anticipatory response.

Water Constraint: Institutional Response

Define shock → Measure impact → Adaptation → Forecast response → **Institutional response** → Firm behavior

1. History of the Water Code

- 1969 Code — water as a *public resource*.
- **1981 Water Code**: **private property**, freely traded for market efficiency.
- *"Paper water"*: limited authority to deny applications ⇒ legal entitlements **exceeded physical supply**.

2. Push towards reform

- 3 pressures: **2010–22 megadrought**; a state **limited reallocation** capability ; **social costs** (wetlands, dispossession).
- 2005: *patente por no uso* — escalating fee on idle rights.
- **2022 Water Code Reform**: perpetual → **30-year revocable** concessions; priority cascade (human above productive use).

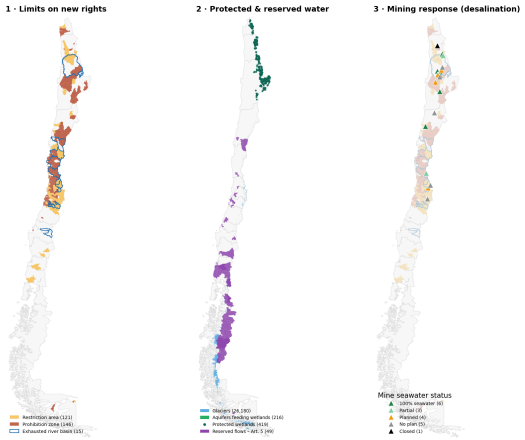
3. Impact on the mining sector

- No law mandates desal — but tighter freshwater limits, stricter **RCA** permitting, DGA oversight & social opposition push mines to **invest in desalination**.
- **16 of 19** mines sit inside / within 10 km of a restricted aquifer or exhausted basin.

Takeaway

The institutional environment **imposes the cost of inaction** on firms still dependent highly on continental freshwater.

Where water is constrained, mines move to the sea



Reading the panels

- **1 — Limits on new rights:** restriction areas, prohibition zones, exhausted river basins.
- **2 — Protected & reserved water:** glaciers, aquifer-feeding & protected wetlands, Article-5 reserved flows.
- **3 — Mining response:** each mine by seawater status (100%, partial, planned, none, closed).

Takeaway

16 of 19 mines sit within **10 km** of a legally restricted, prohibited or protected water source. The rational **long-term firm response** is to switch to a more sustainable water source — **desalination**.

Firm-level evidence — data & approach



Output barely moves — so we look into the operators' own accounts.

Data

- **19 mines** → **14 operators** (consolidated financial entities)
- **Entity** & **Group** financial statements, cost disclosures, annual reports

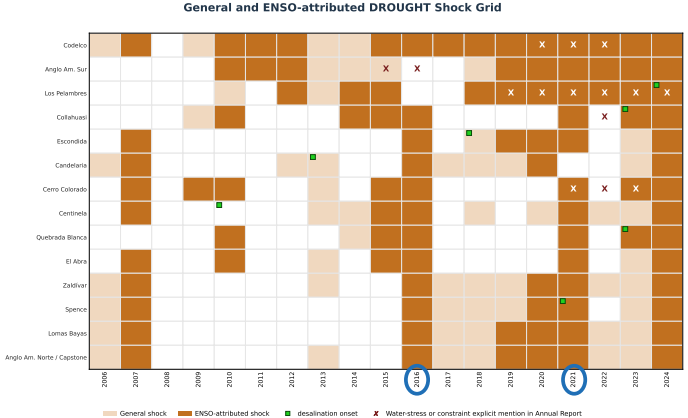
Approach

- Match each operator's firing to **general** and **ENSO-attributed** shocks (**FLOOD** & **DROUGHT**)
- Extract **key financial variables** and **mentions of water-stress**
- Explore **evidence in financial statements in key drought years**

Coverage is uneven across operators and years.

Drought-shock grid: when and where financial stress is disclosed

Define shock → Measure impact → Adaptation → Forecast response → Institutional response → **Firm behavior**



- **2016 & 2021** — drought fires at nearly every operator (**12** and **13** of 14, respectively), at opposite ends of the copper-price cycle.
- **15 mine-years** carry an **explicit mention** of a water-stress event or constraint in the financial reports (across 5 operations).

Source: operator Memorias Anuales (CMF); AR stress flags 2006–2025. Teal row labels = desalinated operations.

Where the drought lands: operating line → balance sheet

Define shock → Measure impact → Adaptation → Forecast response → Institutional response → **Firm behavior**

SHORT TERM · the operating line (cost & production)

2016 price trough · 12/14 operators

Output falls — El Abra −32%, Escondida −13%, Quebrada Blanca −11%, Candelaria −10%.

2021 price peak · 13/14 operators

C1 cost surges — Los Pelambres +10 → 24%, Los Bronces +35%, Collahuasi +43%, Zaldívar +33%.

Output falls — Escondida −16%, Los Pelambres −10%, Zaldívar −9%.

⇒

absorbed by

LONG TERM · adjustment through the balance sheet

Capex & debt build for desalination:

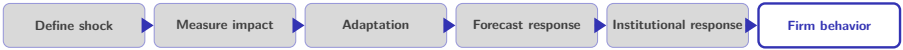
- **Quebrada Blanca** '20→'24: NCL $3.5bn → **$12.0bn**
- **Centinela** '21→'24: debt ∼0 → **$1.9bn** (+$2.5bn 12y financing)
- **Escondida** '15→'17: NCL +$2.4bn → 2018 desal (∼$3.4bn)

⇒ suggests that adaptation to water constraint converts a potentially recurring operating shock into a **debt-financed capital stock**.

We observe patterns of shocks either in **operating flow** (increases in cost and/or reduction in production), and — for operators that commit to desalination — in the **balance sheet** through long-term debt holdings.

C1 = standardised direct cash cost of producing one pound of copper (mining, processing, admin, freight & refining), in US cents/lb. NCL = non-current liabilities (long-term debt).

Two extremes: benchmark and limit case



Escondida — the desalination benchmark

Water source	Desalinated (2018)
ENSO drought years	7
AR water mentions	**0**
Net income (annual)	\$993m–\$4.3bn
Equity	Positive
Going-concern	None
Production	Ongoing

Throughput fell 13% (2016) and 15% (2021) in shock years — attributed to ore grade and mine sequencing, not water availability. No balance-sheet distress marker appears.

Cerro Colorado — provisions, closure & exit

Water source	Continental aquifer
ENSO drought years	7
AR water mentions	3 (2021–23)
IAS 37 provision	\$266m → \$838m
Total equity	\$163m → −\$803m
CAPEX	\$167m → \$9.5m
Production	**Ceased 2023**

Water stress *contributory rather than singular* — low ore grades and weak prices also present; drought foreclosed any extension of mine life.

*IAS 37: international accounting standard requiring firms to book a provision for probable future obligations (here: mine closure & site rehabilitation costs). **Going-concern:** auditor flag raised when a firm's ability to continue operations is in doubt. **CAPEX:** capital expenditure on property, plant & equipment; a collapse signals cessation of reinvestment.*

What we find

The impact on output is channel-specific

- **FLOOD** — the general unusually wet-month on a regional level lowers monthly output ∼**4.7%** (the clean signal).
- **HEAT** — a **null** at monthly resolution (routine exposure the north operates around).
- **DROUGHT** — registers *only* as **ENSO-consistent persistent** water stress (−0.277), coincided with the **post-2020 megadrought**.

As the sector absorbed the shock

- Mines do **not** time production to ENSO forecasts — they adapt **structurally**, via multi-year desalination programmes.
- The cost surfaces off the output statement: **rising cash costs, capex, closure provisions**, and one terminal closure.
- Desalinated operations show almost none of it.

The contribution

We studied the reaction of Chilean copper mining sector on climate shocks — including ENSO-attributed — **at the level of the individual mine**, explored the application of ENSO-forecastability and **document** evidence of adaptation to the water-scarcity through institutional response (water-rights management), as well as through firm-level investment and operational decisions. *Adaptation here is a complex ongoing story — ENSO is only part of it.*

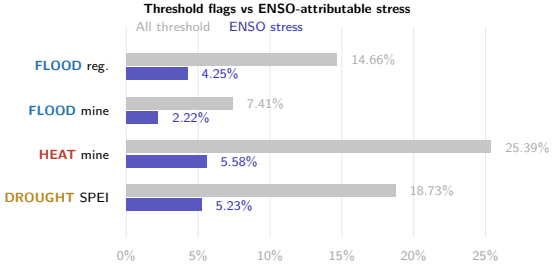
Thank you

Questions & discussion

Appendix — how often does the stress flag fire?

Channel	All threshold	ENSO stress
FLOOD regional mean	2,206 (14.66%)	639 (4.25%)
FLOOD mine-cell	1,114 (7.41%)	334 (2.22%)
HEAT mine-cell	3,816 (25.39%)	838 (5.58%)
DROUGHT SPEI-12	2,780 (18.73%)	776 (5.23%)

- Full master ledger (1960–2025); denominators are valid channel mine-months.
- Active regression firings narrower: FLOOD reg. 602/115, FLOOD mine 293/65, HEAT 1,127/310 (Spec A/B).
- DROUGHT annualised: 776 SPEI-12 months → lagged share in 369 mine-year panel.



Spec B is intentionally selective: stress = threshold crossed × active, directionally consistent ENSO gate.

Channel-specific inference: **FLOOD** has broad wet-month exposure for the headline effect, **HEAT** mostly marks routine exposure, and **DROUGHT** is persistent rather than a discrete monthly shock.

Appendix — firm-level responses to climate stress

*Two horizons of response. The **DROUGHT** long-term strategic cell — **desalination** — is the response we measure.*

	FLOOD	HEAT	DROUGHT
Hazard	Flooded pits & roads, washed-out access/ports, tailings-dam overflow	Worker heat stress, equipment overheating, dust & evaporation in arid sites	Freshwater scarcity & curtailed rights: throughput cuts, forced closure. **Water is a key input**
Acute *days–weeks*	Early-warning & stormwater protocols; halt ops, secure tailings, reroute logistics	Heat-stress protocols (DS 594): shift timing, hydration — little output adjustment	Curtail throughput, prioritise high-grade ore, truck in water; closure if rights restricted
Strategic *years · capital*	Drainage/freeboard, dry-stack tailings (GISTM), hardened logistics	Automation/remote ops, equipment cooling, workforce acclimatisation	**Switch to desalinated seawater** + pipelines, recycling, new rights — **the dominant capital response**

Appendix — Spence within-mine event study

Channel	n stress pre	n stress post	β_1 pre (p)	marg. post
HEAT mine-cell	42	13	−0.18 (**0.03**)	−0.03 (β_2: p=0.10)
FLOOD region-mean	4	0	—	*not estimable*
DROUGHT	—	—	—	*not enough post-switch window*

HEAT — cushioned ✓

Pre-desalination heat stress cut output ∼18% (p=0.03). After SGO commissioned on desalinated seawater (2021), the same heat event has ≈ **zero** effect.

FLOOD region-mean — not testable

Spence has **4 pre / 0 post** qualifying flood months — unable to compare pre vs. post.

Caveat: the 2021 commissioning bundles desalination with a new concentrator and process control, and heat has no water mechanism — read as suggestive support from one mine, not identification.